$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{\Upsilon_1 k^2}{2} & \frac{k^2 \overset{(4)}{\kappa}_4}{2\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{k^2 \overset{(4)}{\kappa}_4}{2\sqrt{2}} & \frac{\Upsilon_2 k^2}{2} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{3 k^2 \overset{(4)}{\kappa}_1}{2} & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{\Upsilon_2 k^2}{2\,\mathrm{Det}(1^+)} & -\frac{k^2 \overset{(4)}{\kappa}_4}{2\sqrt{2}\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{k^2 \overset{(4)}{\kappa}_4}{2\sqrt{2}\,\mathrm{Det}(1^+)} & -\frac{\Upsilon_1 k^2}{2\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{2}{3 k^2 \overset{(4)}{\kappa}_1} & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

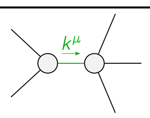
Abbreviations used in matrices

$$\Upsilon_1 == 3\overset{(4)}{\kappa}_1 + 2\overset{(4)}{\kappa}_3 \;\&\&\; \Upsilon_2 == \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \;\&\&\; \mathrm{Det}(1^+) == -\frac{1}{8} k^4 \left(6\overset{(4)}{\kappa}_1 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2\right)$$

Lagrangian

$$\overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta} \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} + \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^{\delta} -$$
$$3\overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} + \frac{1}{2}\overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{0^-}{}^{\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta \mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#2}_{1^-}{}^{\alpha} == 0$	3	$\partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$
$\mathcal{J}^{\#1}_{1^-}{}^{\alpha} == 0$	3	$\partial_\chi\partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#1}_{2^-}{}^{\alpha\beta\chi} == 0$	5	$3\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\alpha}{}_{\delta} == 3\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	13	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{24\overset{(4)}{\kappa}_1{}^2 + 8\overset{(4)}{\kappa}_1 (2\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2}{\overset{(4)}{\kappa}_1 (6\overset{(4)}{\kappa}_1 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)}$

Resolved unitarity condition(s):

$$\Big(\overset{(4)}{\kappa}_4 < 0 \;\&\&\; \Big(\Big(\overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; 0 < \overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3{}^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4{}^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4}\Big) \;||\; \Big(-\frac{\overset{(4)}{\kappa}_4}{2} < \overset{(4)}{\kappa}_3 < -\overset{(4)}{\kappa}_4 \;\&\&\; 0 < \overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3{}^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4{}^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4}\Big) \;||$$

$$\Big(\overset{(4)}{\kappa}_3 == -\overset{(4)}{\kappa}_4 \;\&\&\; \overset{(4)}{\kappa}_1 > 0\Big) \;||\; \Big(\overset{(4)}{\kappa}_3 > -\overset{(4)}{\kappa}_4 \;\&\&\; \Big(\overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3{}^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4{}^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4} \;||\; \overset{(4)}{\kappa}_1 > 0\Big)\Big)\Big)\Big) \;||$$

$$\Big(\overset{(4)}{\kappa}_4 == 0 \;\&\&\; \Big(\Big(\overset{(4)}{\kappa}_3 < 0 \;\&\&\; 0 < \overset{(4)}{\kappa}_1 < -\frac{2\overset{(4)}{\kappa}_3}{3}\Big) \;||\; \Big(\overset{(4)}{\kappa}_3 > 0 \;\&\&\; \Big(\overset{(4)}{\kappa}_1 < -\frac{2\overset{(4)}{\kappa}_3}{3} \;||\; \overset{(4)}{\kappa}_1 > 0\Big)\Big)\Big)\Big) \;||$$

$$\Big(\overset{(4)}{\kappa}_4 > 0 \;\&\&\; \Big(\Big(\overset{(4)}{\kappa}_3 < -\overset{(4)}{\kappa}_4 \;\&\&\; 0 < \overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3{}^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4{}^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4}\Big) \;||$$

$$\Big(\overset{(4)}{\kappa}_3 == -\overset{(4)}{\kappa}_4 \;\&\&\; \overset{(4)}{\kappa}_1 > 0\Big) \;||\; \Big(-\overset{(4)}{\kappa}_4 < \overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; \Big(\overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3{}^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4{}^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4} \;||\; \overset{(4)}{\kappa}_1 > 0\Big)\Big) \;||$$

$$\Big(\overset{(4)}{\kappa}_3 == -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; \Big(\overset{(4)}{\kappa}_1 < 0 \;||\; \overset{(4)}{\kappa}_1 > 0\Big)\Big) \;||\; \Big(\overset{(4)}{\kappa}_3 > -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; \Big(\overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3{}^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4{}^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4} \;||\; \overset{(4)}{\kappa}_1 > 0\Big)\Big)\Big)\Big)$$